

Weilin Li

Last updated: August 2026

Contact Information

Office: Room 308B, Marshak Science Building
City College of New York
160 Convent Ave, New York, NY 10031

Email: wli6@ccny.cuny.edu
Website: <https://weilinlimath.github.io/>
[Google Scholar Profile](#)

Research Interests

Applied analysis, computational mathematics, signal processing, machine learning

Employment

City University of New York

Doctoral Faculty, Graduate Center

2025–

Assistant Professor, City College of New York

2022–

New York University

Visiting Assistant Professor

2021–22

Courant Instructor

2018–21

Education

University of Maryland, College Park

2018

Ph.D. Mathematics

Thesis: Topics in Harmonic Analysis, Sparse Representations, and Data Analysis

Advisors: John J. Benedetto and Wojciech Czaja

Cornell University

2013

B.A. Mathematics, Summa Cum Laude

Thesis: Boundary Value Problems on a Half-Sierpinski Gasket

Advisor: Robert S. Strichartz

Preprints

4. Hana Huber and Weilin Li. A sharp analysis of Root-MUSIC: locations of correct and extraneous roots. [\[arXiv\]](#)
3. Albert Fannjiang and Weilin Li. Multidimensional Gradient-MUSIC: A global nonconvex optimization framework for optimal resolution. [\[arXiv\]](#)
2. C. Sinan Güntürk and Weilin Li. Approximation of functions with one-bit neural networks. [\[arXiv\]](#)
1. Albert Fannjiang and Weilin Li. Optimal structured approximation of Fourier subspaces, Toeplitz matrices, and exponential sums. [\[arXiv\]](#)

Journal Publications

19. Albert Fannjiang, Weilin Li, and Wenjing Liao. Optimality of Gradient-MUSIC for spectral estimation. *Information and Inference*, vol. 15, no. 3, 2026. [\[journal\]](#)
18. Weilin Li, Karl Otness, Kui Ren, and Donsub Rim. An explicit spectral decomposition of the ADRT. *BIT Numerical Mathematics*, vol. 66, no. 37, 2026. [\[journal\]](#)
17. Oleg Asipchuk, Laura De Carli, Weilin Li. Concerning the stability of exponential systems and Fourier matrices. *Journal of Fourier Analysis and Applications*, vol. 31, no. 67, 2025. [\[journal\]](#)

16. Weilin Li. Nonharmonic multivariate Fourier transforms and matrices: condition numbers and hyperplane geometry. *Applied and Computational Harmonic Analysis*, vol. 79, pp. 101791, 2025. [\[journal\]](#)
15. Sjoerd Dirksen, Weilin Li, and Johannes Maly. Subspace and DOA estimation under coarse quantization. *IEEE Transactions on Information Theory*, vol. 71, no. 10, pp. 8149–8168, 2025. [\[journal\]](#)
14. Weilin Li. Multiscale estimates for the condition number of non-harmonic Fourier matrices. *Mathematics of Computation*, vol. 94, no. 356, pp. 2895–2929, 2025. [\[journal\]](#)
13. C. Sinan Güntürk and Weilin Li. Approximation with one-bit polynomials in Bernstein form. *Constructive Approximation (special issue for Ron DeVore’s 80th birthday)*, vol. 57, no. 2, pp. 601–630, 2023. [\[journal\]](#)
12. Weilin Li, Kui Ren, and Donsub Rim. A range characterization of the single-quadrant ADRT. *Mathematics of Computation*, vol. 92, no. 339, pp. 283–306, 2023. [\[journal\]](#)
11. C. Sinan Güntürk and Weilin Li. Quantization for spectral super-resolution. *Constructive Approximation*, vol. 56, no. 3, pp. 619–648, 2022. [\[journal\]](#)
10. Weilin Li, Zengying Zhu, Weiguo Gao, and Wenjing Liao. Stability and super-resolution of MUSIC and ESPRIT for multi-snapshot spectral estimation. *IEEE Transactions on Signal Processing*, vol. 70, pp. 4555–4570, 2022. [\[journal\]](#)
9. Ilya Kavalierov, Weilin Li, Wojciech Czaja, and Rama Chellappa. 3-D Fourier scattering transform and classification of hyperspectral images. *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 12, pp. 10312–10327, 2021. [\[journal\]](#)
8. Weilin Li and Wenjing Liao. Stable super-resolution limit and smallest singular value of restricted Fourier matrices. *Applied and Computational Harmonic Analysis*, vol. 51, no. 1, pp. 118–156, 2021. [\[journal\]](#)
7. Weilin Li. Generalization error of minimum weighted norm and kernel interpolation. *SIAM Journal on Mathematics of Data Science*, vol. 3, no. 1, pp. 414–438, 2021. [\[journal\]](#)
6. Weilin Li, Wenjing Liao, and Albert Fannjiang. Super-resolution limit of the ESPRIT algorithm. *IEEE Transactions on Information Theory*, vol. 66, no. 7, pp. 4593–4608, 2020. [\[journal\]](#)
5. Wojciech Czaja and Weilin Li. Rotationally invariant time-frequency scattering transforms. *Journal of Fourier Analysis and Applications*, vol. 26, no. 1, pp. 1–23, 2020. [\[journal\]](#)
4. John J. Benedetto and Weilin Li. Super-resolution by means of Beurling minimal extrapolation. *Applied and Computational Harmonic Analysis*, vol. 48, no. 1, pp. 218–241, 2020. [\[journal\]](#)
3. Wojciech Czaja and Weilin Li. Analysis of time-frequency scattering transforms. *Applied and Computational Harmonic Analysis*, vol. 47, no. 1, pp. 149–171, 2019. [\[journal\]](#)
2. John Peterson, Weilin Li, Brian Cesar-Tondreau, John Bird, Kevin Kochersberger, Wojciech Czaja, and Morgan McLean. Experiments in unmanned aerial vehicle/unmanned ground vehicle radiation search. *Journal of Field Robotics*, vol. 36, no. 4, pp. 818–845, 2019. [\[journal\]](#)
1. Weilin Li and Robert S. Strichartz. Boundary value problems on a half Sierpinski gasket. *Journal of Fractal Geometry*, vol. 1, no. 1, pp. 1–43, 2014. [\[journal\]](#)

External Grants

3. *National Science Foundation, Division of Mathematical Sciences, Sole-PI Grant* 2026–
A sole-PI project “A rigorous optimization framework for high-resolution parameter estimation on the continuum”, awarded by the NSF Computational Mathematics program with a budget of \$215,000.
2. *National Science Foundation, Division of Mathematical Sciences, Sole-PI Grant* 2023–2026
A sole-PI project “Multidimensional and compressive super-resolution: theory, computation, and

fundamental limits”, awarded by the NSF Computational Mathematics program with a budget of \$176,476.

1. *AMS-Simons Travel Grant* 2019–22
Awarded \$5,000 for travel, funded by the AMS-Simons Travel Grants Program.

Internal Grants

5. *PSC-CUNY Track I Grant* 2026–
4. *PSC-CUNY Traditional B Grant* 2025–26
3. *PSC-CUNY Traditional B Grant* 2024–25
2. *PSC-CUNY Traditional B Grant* 2023–24
1. *Cohen Foundation Research Support, New York University* 2018–19

Awards and Honors

5. *Charles Chui Young Researcher Best Paper Award* 2021
for “Stable super-resolution limit and smallest singular value of restricted Fourier matrices”
4. *James C. Alexander Prize for Graduate Research, University of Maryland* 2018
One of two PhD students who won this prize for outstanding thesis.
3. *Program Associate, Mathematical Sciences Research Institute* 2017
Participated as a student visitor for the 2017 Harmonic Analysis program at Mathematical Sciences Research Institute, Berkeley California.
2. *Ann G. Wylie Dissertation Fellowship, University of Maryland* 2017
Competitive university fellowship that was awarded to 50 graduate students in 2017.
1. *Graduated Summa Cum Laude in Mathematics, Cornell University* 2013
Graduated with highest honors in mathematics from Cornell University.

Recorded Online Talks

3. [“A nonconvex optimization approach to spectral estimation”](#) 2026
1W-MINDS: One World Mathematics of INformation, Data, and Signals
2. [“Function approximation with one-bit Bernstein and neural networks”](#) 2021
Faraway Fourier Talks
1. [“Super-resolution, subspace methods, and Fourier matrices”](#) 2021
1W-MINDS: One World Mathematics of INformation, Data, and Signals

Conference Presentations

19. 15th International Conference on Sampling Theory and Applications, Vienna, Austria 2025
18. Frame Theory Days, St. Louis University, MO 2024
17. NSF PI Meeting, University of Washington Seattle, WA 2024
16. SIAM Conference on Applied Linear Algebra, Paris, France 2024
15. AMS Spring Sectional Meeting, Tallahassee, FL 2024
14. 11th Applied Inverse Problems Conference, Göttingen, Germany 2023
13. 17th US National Congress on Computational Mechanics, Albuquerque, New Mexico 2023
12. TUM Science and Study Center Raitenhaslach, Germany 2023
11. Fall Fourier Talks, University of Maryland, College Park, MD 2022

10. Focus Program, Fields Institute, Toronto, Canada	2022
9. Joint Math Meetings, Seattle, WA,	2022
8. SIAM Conference on Imaging Science, Berlin, Germany	2022
7. 11th WHISPERS, Amsterdam, Netherlands (online)	2021
6. Randomness and Determinism in Compressive Data Acquisition, Texas A&M University	2019
5. AMS Fall Western Sectional Meeting, San Francisco State University, CA	2018
4. 7th International Conference on Computational Harmonic Analysis, Vanderbilt University	2018
3. AMS Sectional Meeting, University of Florida, Orlando, FL	2017
2. Joint Mathematics Meetings, Baltimore, MD	2014
1. Joint Mathematics Meetings, San Diego, CA	2013

Seminar Talks

29. CUNY Harmonic Analysis and PDE seminar, CUNY Graduate Center, New York	2025
28. Colloquium, New Jersey Institute of Technology, Newark	2025
27. Analysis and Data Science Seminar, State University of New York, Albany	2024
26. Data-Enabled Science Seminar, University of Houston	2024
25. Applied and Computational Math Seminar, Georgia Institute of Technology	2024
24. Colloquium, Florida International University	2023
23. Machine Learning Seminar, University of Massachusetts at Amherst	2023
22. Colloquium, The City College of New York	2022
21. Applied Math Colloquium, Columbia University	2022
20. Harmonic Analysis and PDE Seminar, Graduate Center of CUNY (online)	2022
19. Applied/PDE/Data Seminar, University of California, Santa Barbra (online)	2021
18. Faraway Fourier Talks (online)	2021
17. 1W-MINDS: One World Mathematics of INformation, Data, and Signals (online)	2021
16. Scientific Computing And Numerics, Cornell University (online)	2021
15. Computational and Applied Math Seminar, Tufts University	2021
14. Computational Analysis Seminar, Vanderbilt University (online)	2021
13. Colloquium, Washington University in St. Louis (online)	2021
12. Colloquium, North Carolina State University (online)	2020
11. Seminar in Mathematical Physics/PDE, University of California, San Diego	2019
10. Applied and Computational Math Seminar, University of California, Irvine	2019
9. Norbert Wiener Center Seminar, University of Maryland, College Park	2019
8. Analysis Seminar, Courant Institute of Mathematical Sciences	2019
7. Norbert Wiener Center Seminar, University of Maryland, College Park	2018
6. Applied Math and Analysis Seminar, Duke University	2018
5. Harmonic Analysis and Signal Processing Seminar, Courant Institute	2017
4. Applied and Computational Mathematics Seminar, Georgia Institute of Technology	2017
3. Norbert Wiener Center Seminar, University of Maryland, College Park	2017
2. Graduate Student Seminar, Mathematical Sciences Research Institute, Berkeley	2017
1. Research Interaction Team on Deep Learning, University of Maryland, College Park	2016

Courses Taught

City College of New York: (* indicates a graduate level class)

- *Math 434/A34 Theory of Functions of a Real Variables (Fa23, Sp25)
- *Math 432/A32 Theory of Functions of a Complex Variable (Fa25)
- Math 377 Applied Statistics and Probability (Sp23, Sp24)
- Math 328 Methods of Numerical Analysis (Fa24)
- Math 324 Advanced Calculus II (Fa22)

New York University: (* indicates a graduate level class)

- *Math 2110 Linear Algebra I (Fa21)
- *Math 2111 Linear Algebra (Fa21)
- Math 325 Analysis (Sp22, Sp21, Sp20)
- Math 211 Math for Economics I (Fa19)
- Math 123 Calculus III (Sp19)
- Math 122 Calculus II (Fa20)
- Math 121 Calculus I (Fa18)